AWS Monitoring, Audit and Performance

CloudWatch, CloudTrail & AWS Config

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Amazon CloudWatch Metrics



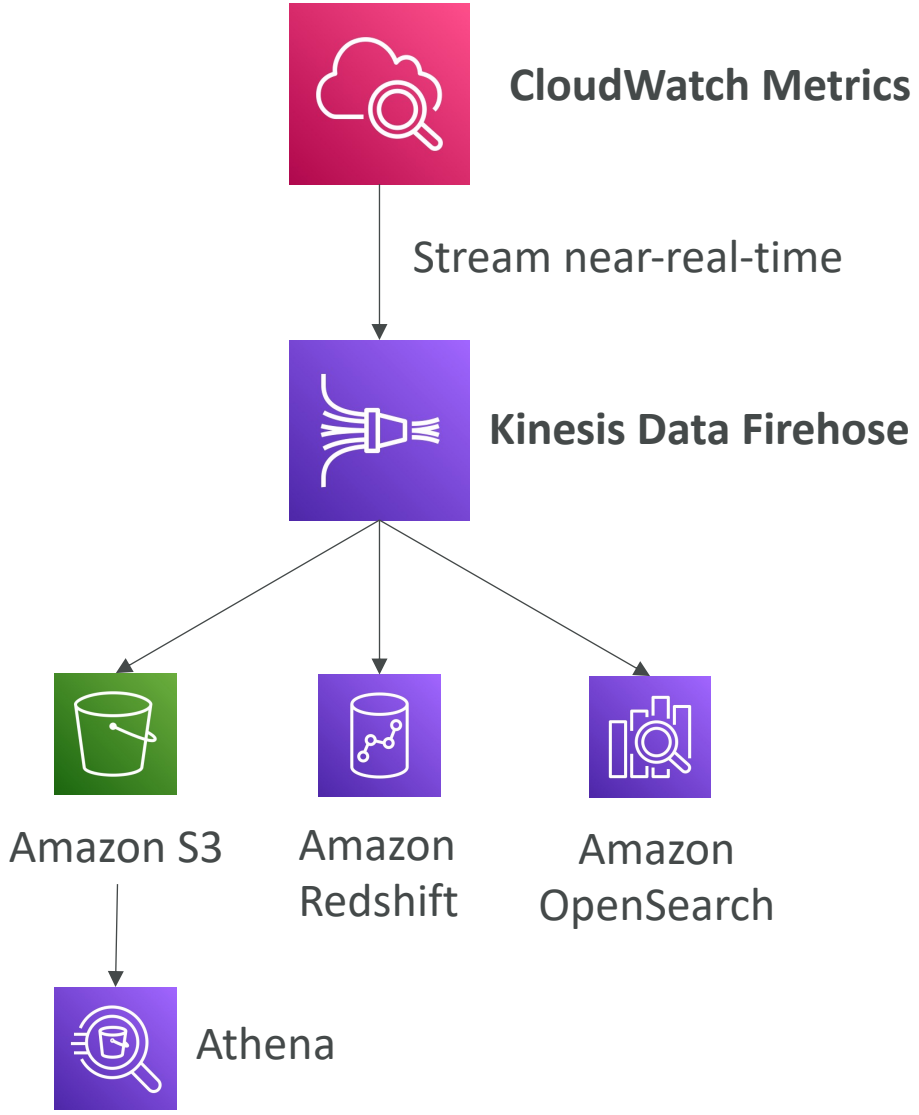
- CloudWatch provides metrics for *every* service in AWS
- **Metric** is a variable to monitor (CPUUtilization, NetworkIn…)
- Metrics belong to **namespaces**
- **Dimension** is an attribute of a metric (instance id, environment, etc…).
- Up to 30 dimensions per metric
- Metrics have **timestamps**
- Can create CloudWatch dashboards of metrics
- Can create **CloudWatch Custom Metrics** (for the RAM for example)

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CloudWatch Metric Streams

- Continually stream CloudWatch metrics to a destination of your choice, with **near-real-time delivery** and low latency.
  - Amazon Kinesis Data Firehose (and then its destinations)
  - 3rd party service provider: Datadog, Dynatrace, New Relic, Splunk, Sumo Logic…
- Option to **filter metrics** to only stream a subset of them



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CloudWatch Logs



- **Log groups:** arbitrary name, usually representing an application
- **Log stream:** instances within application / log files / containers
- Can define log expiration policies (never expire, 1 day to 10 years…)
- **CloudWatch Logs can send logs to:**
  - Amazon S3 (exports)
  - Kinesis Data Streams
  - Kinesis Data Firehose
  - AWS Lambda
  - OpenSearch
- Logs are encrypted by default
- Can setup KMS-based encryption with your own keys

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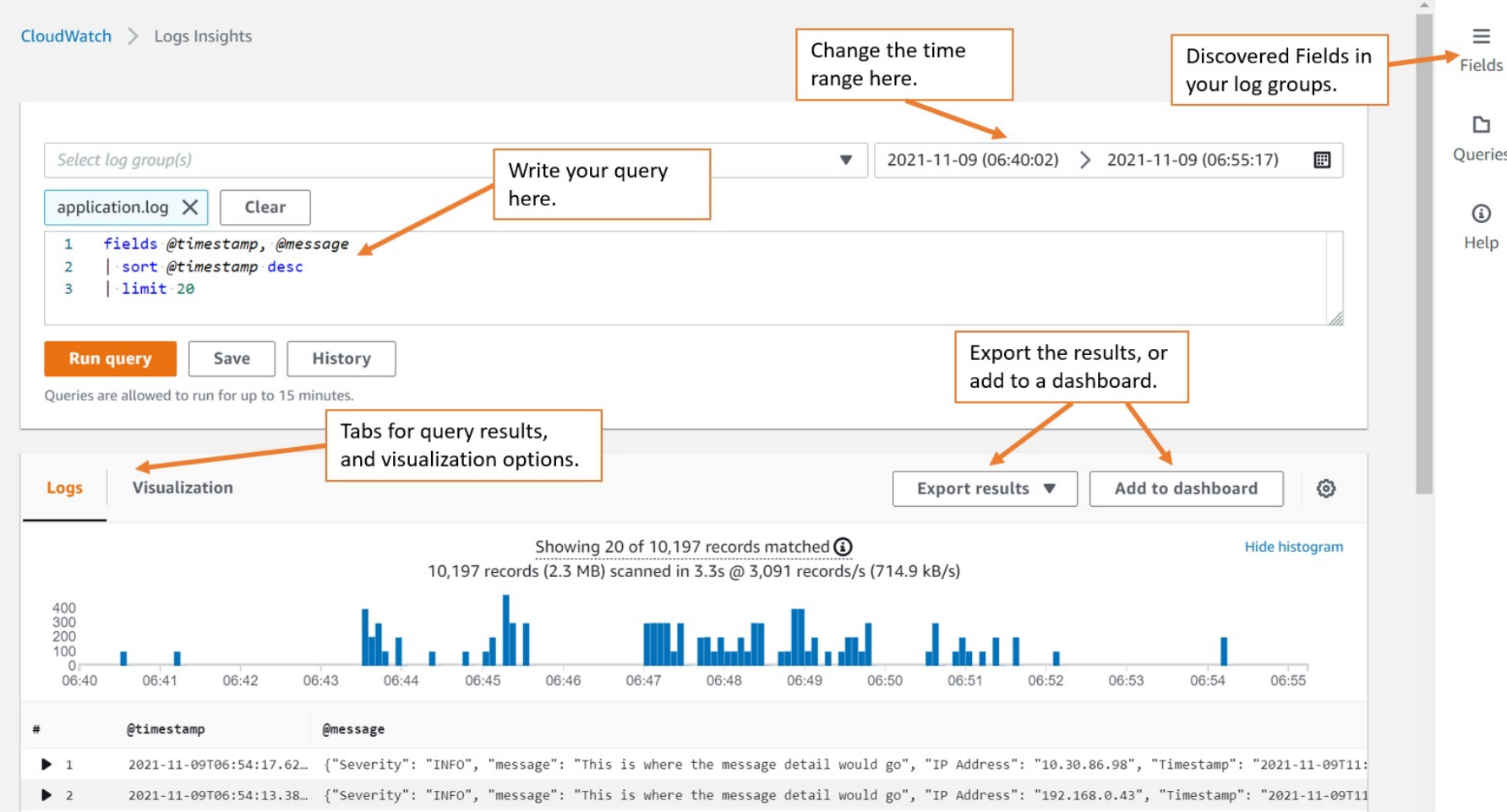
CloudWatch Logs - Sources

- SDK, CloudWatch Logs Agent, CloudWatch Unified Agent
- Elastic Beanstalk: collection of logs from application
- ECS: collection from containers
- AWS Lambda: collection from function logs
- VPC Flow Logs: VPC specific logs
- API Gateway
- CloudTrail based on filter
- Route53: Log DNS queries

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CloudWatch Logs Insights



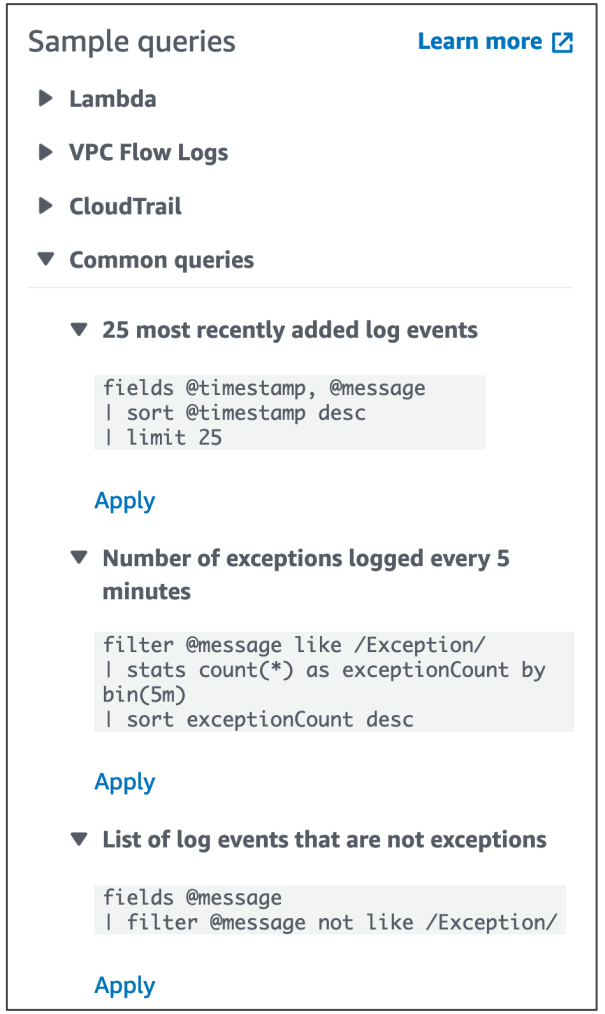
https://mng.workshop.aws/operations-2022/detect/cwlogs.html

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CloudWatch Logs Insights

- Search and analyze log data stored in CloudWatch Logs
- Example: find a specific IP inside a log, count occurrences of "ERROR" in your logs…
- Provides a purpose-built query language
  - Automatically discovers fields from AWS services and JSON log events
  - Fetch desired event fields, filter based on conditions, calculate aggregate statistics, sort events, limit number of events…
  - Can save queries and add them to CloudWatch Dashboards
- Can query multiple Log Groups in different AWS accounts
- It's a query engine, not a real-time engine



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CloudWatch Logs – S3 Export



- Log data can take **up to 12 hours** to become available for export
- The API call is **CreateExportTask**

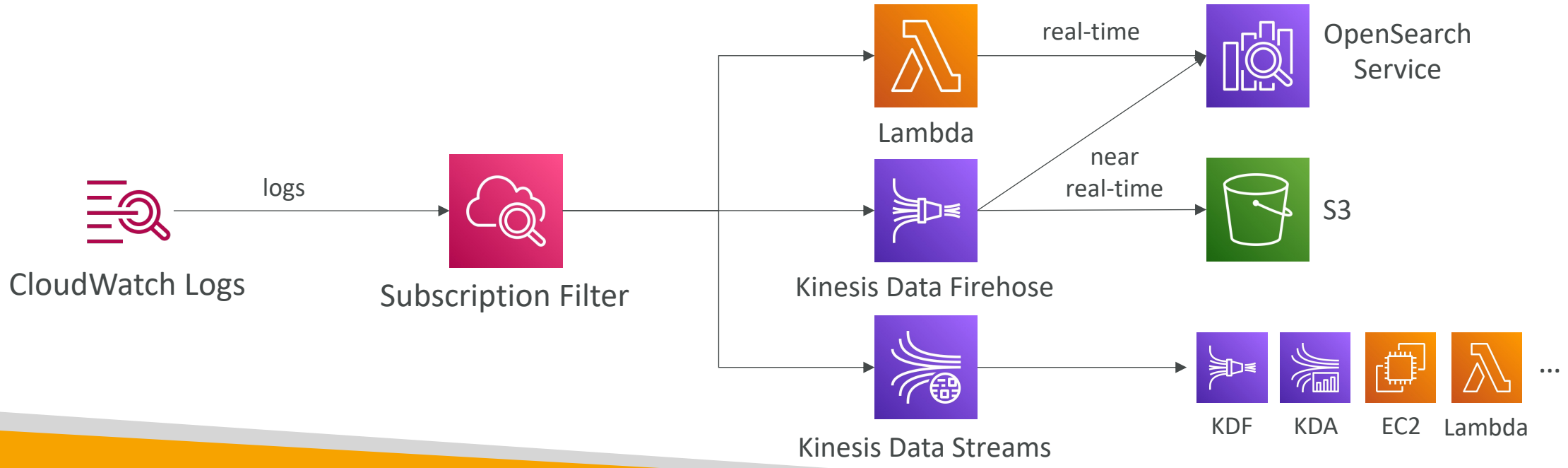
- Not near-real time or real-time… use Logs Subscriptions instead

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CloudWatch Logs Subscriptions

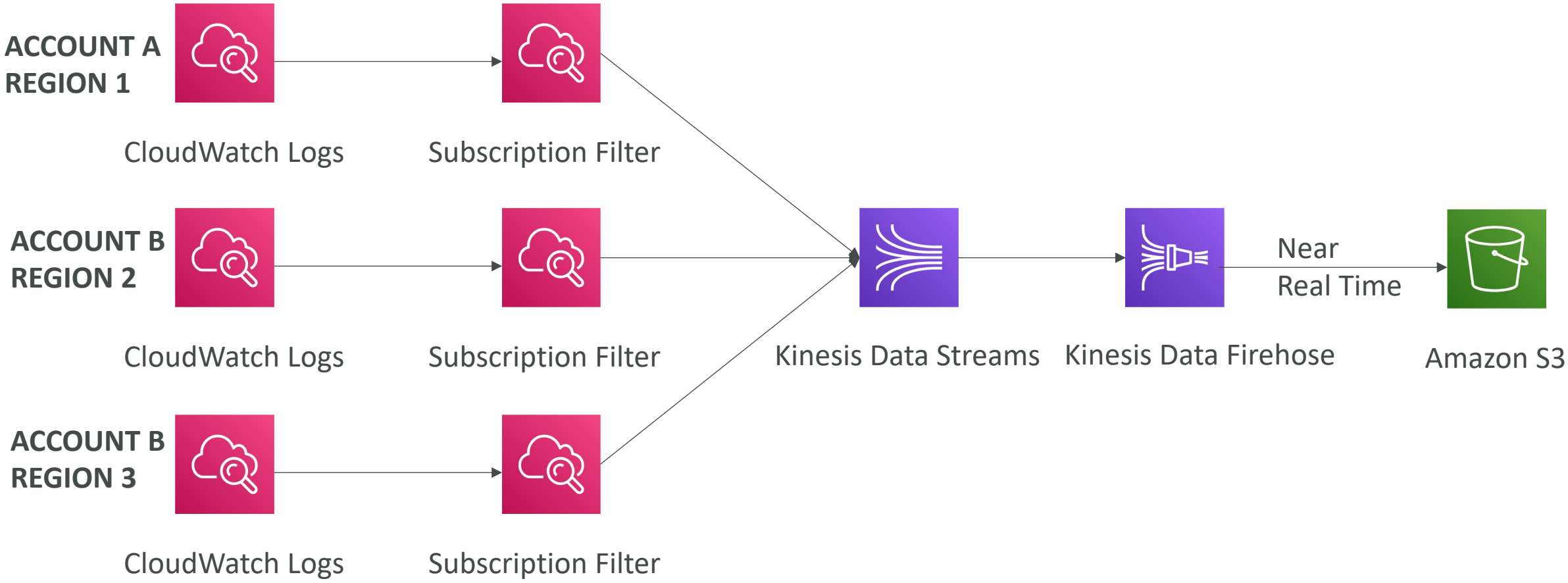
- Get a real-time log events from CloudWatch Logs for processing and analysis
- Send to Kinesis Data Streams, Kinesis Data Firehose, or Lambda
- **Subscription Filter** – filter which logs are events delivered to your destination



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CloudWatch Logs Aggregation Multi-Account & Multi Region

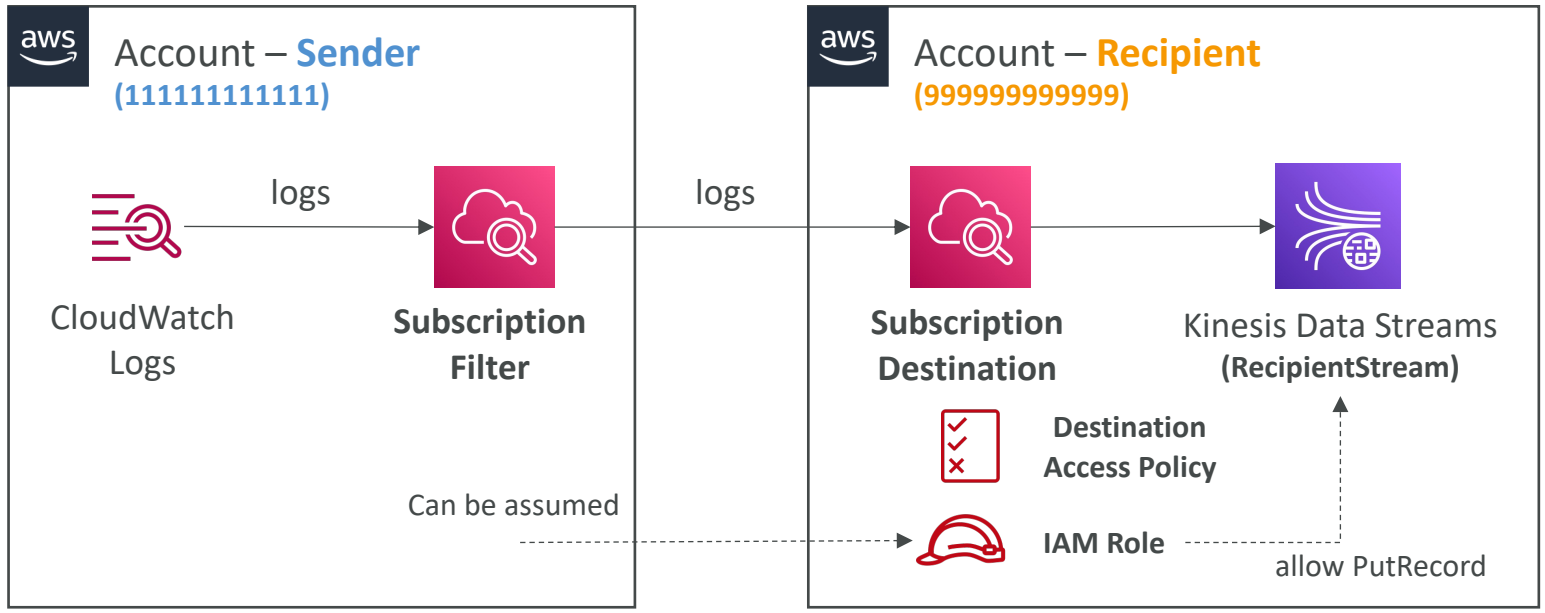


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CloudWatch Logs Subscriptions

- **Cross-Account Subscription** – send log events to resources in a different AWS account (KDS, KDF)



IAM Role (Cross-Account)

```
{
  "Statement": [
    {
      "Effect": "Allow",
      "Action": "kinesis:PutRecord",
      "Resource": "arn:aws:kinesis:us-east-1:
999999999999:stream/RecipientStream"
    }
  ]
}
```

Destination Access Policy

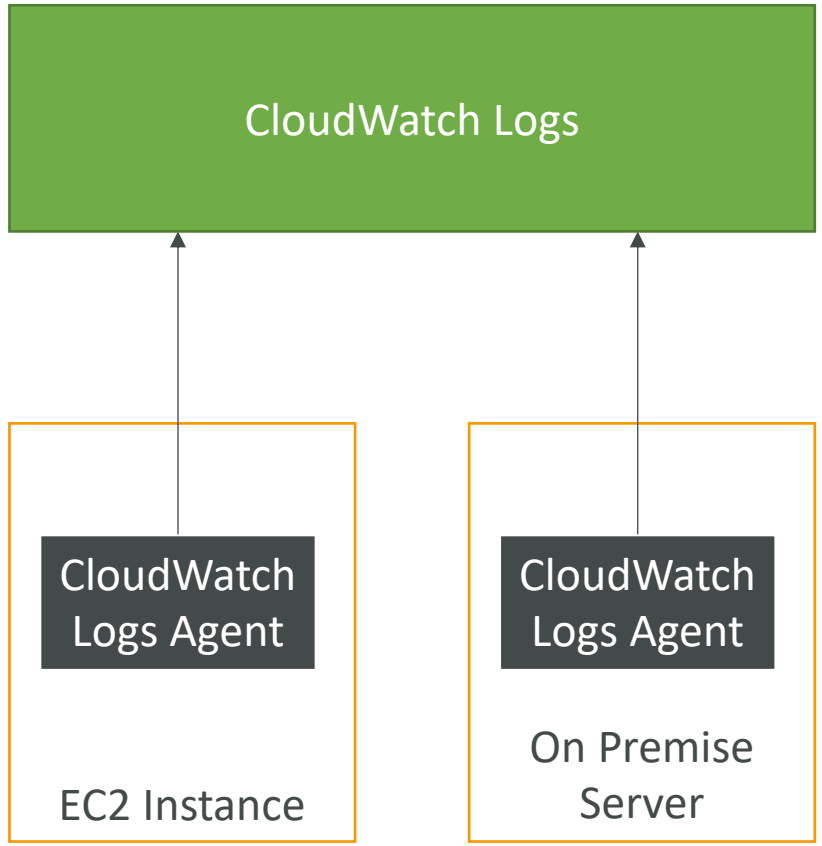
```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "",
      "Effect": "Allow",
      "Principal": {
        "AWS": "111111111111"
      },
      "Action": "logs:PutSubscriptionFilter",
      "Resource": "arn:aws:logs:us-east-1:999999999999:
destination:testDestination"
    }
  ]
}
```

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CloudWatch Logs for EC2

- By default, no logs from your EC2 machine will go to CloudWatch
- You need to run a CloudWatch agent on EC2 to push the log files you want
- Make sure IAM permissions are correct
- The CloudWatch log agent can be setup on-premises too



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CloudWatch Logs Agent & Unified Agent

- For virtual servers (EC2 instances, on-premises servers…)
- **CloudWatch Logs Agent**
  - Old version of the agent
  - Can only send to CloudWatch Logs
- **CloudWatch Unified Agent**
  - Collect additional system-level metrics such as RAM, processes, etc…
  - Collect logs to send to CloudWatch Logs
  - Centralized configuration using SSM Parameter Store

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CloudWatch Unified Agent – Metrics

- Collected directly on your Linux server / EC2 instance

- **CPU** (active, guest, idle, system, user, steal)
- **Disk metrics** (free, used, total), Disk IO (writes, reads, bytes, iops)
- **RAM** (free, inactive, used, total, cached)
- **Netstat** (number of TCP and UDP connections, net packets, bytes)
- **Processes** (total, dead, bloqued, idle, running, sleep)
- **Swap Space** (free, used, used %)

- Reminder: out-of-the box metrics for EC2 – disk, CPU, network (high level)

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CloudWatch Alarms



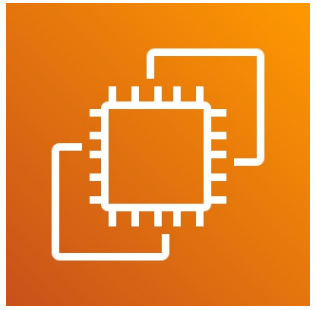
- Alarms are used to trigger notifications for any metric
- Various options (sampling, %, max, min, etc…)
- Alarm States:
  - OK
  - INSUFFICIENT_DATA
  - ALARM
- Period:
  - Length of time in seconds to evaluate the metric
  - High resolution custom metrics: 10 sec, 30 sec or multiples of 60 sec

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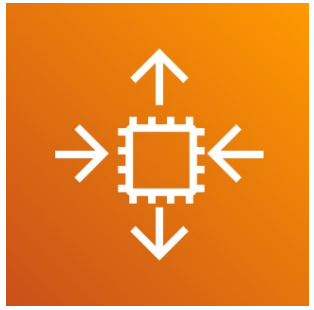
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CloudWatch Alarm Targets

- Stop, Terminate, Reboot, or Recover an EC2 Instance
- Trigger Auto Scaling Action
- Send notification to SNS (from which you can do pretty much anything)



Amazon EC2



EC2 Auto Scaling



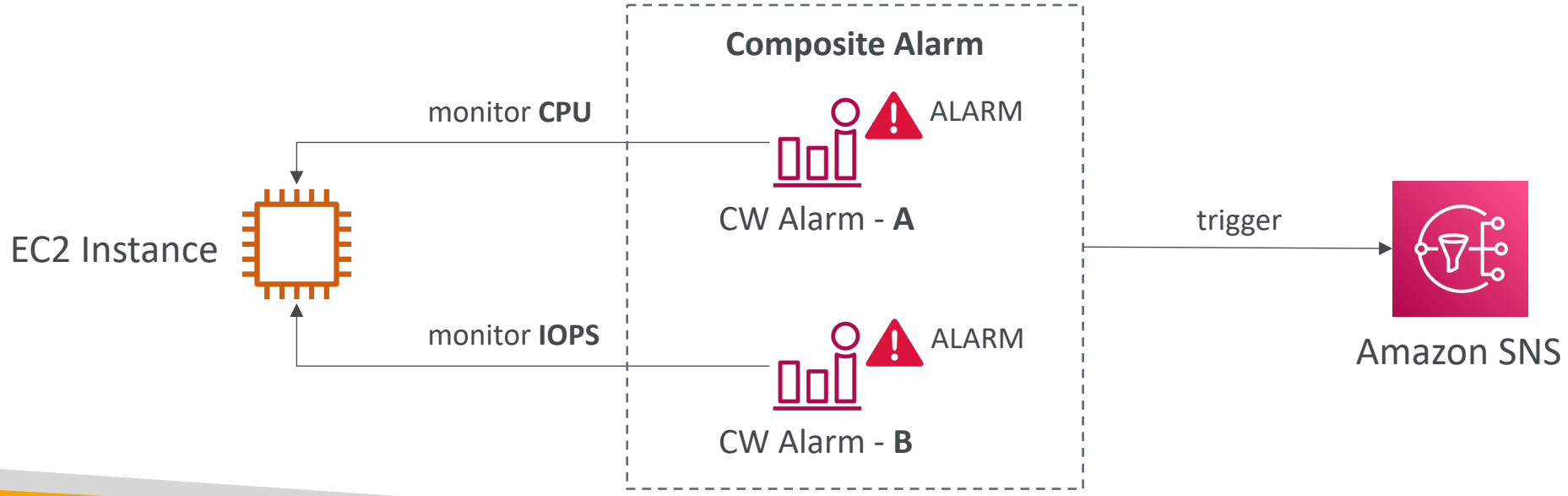
Amazon SNS

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CloudWatch Alarms – Composite Alarms

- CloudWatch Alarms are on a single metric
- **Composite Alarms are monitoring the states of multiple other alarms**
- AND and OR conditions
- Helpful to reduce "alarm noise" by creating complex composite alarms

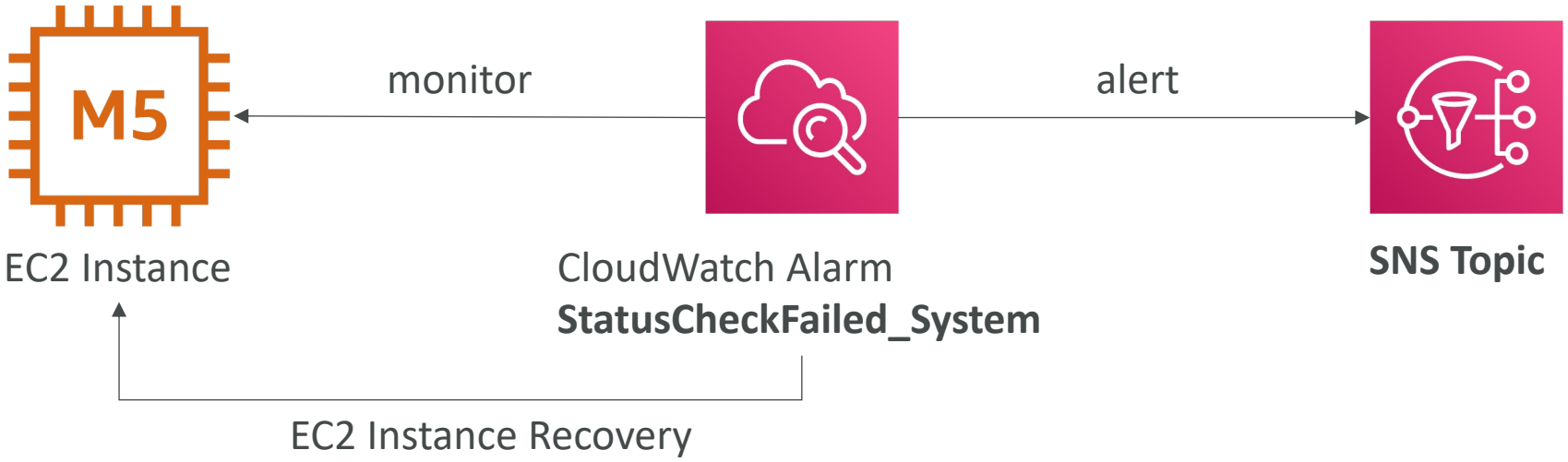


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EC2 Instance Recovery

- Status Check:
  - Instance status = check the EC2 VM
  - System status = check the underlying hardware
  - Attached EBS status = check attached EBS volumes



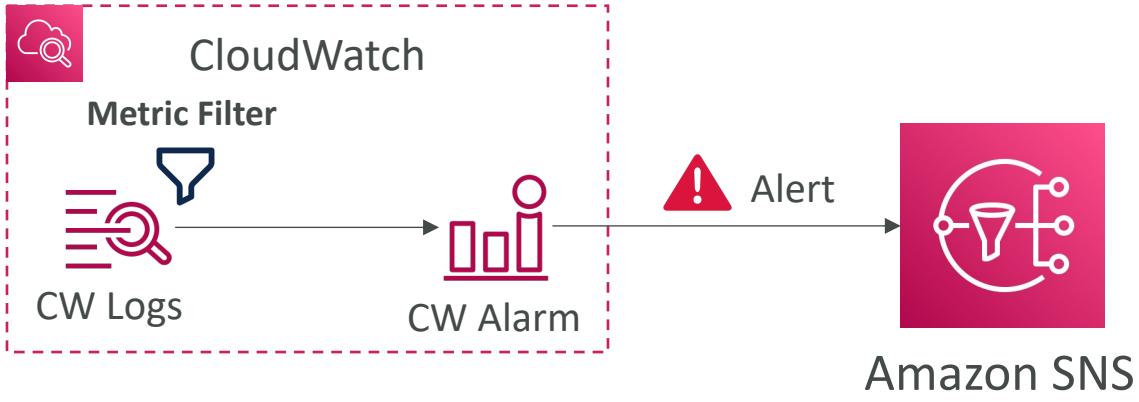
- **Recovery:** Same Private, Public, Elastic IP, metadata, placement group

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CloudWatch Alarm: good to know

- Alarms can be created based on CloudWatch Logs Metrics Filters



- To test alarms and notifications, set the alarm state to Alarm using CLI

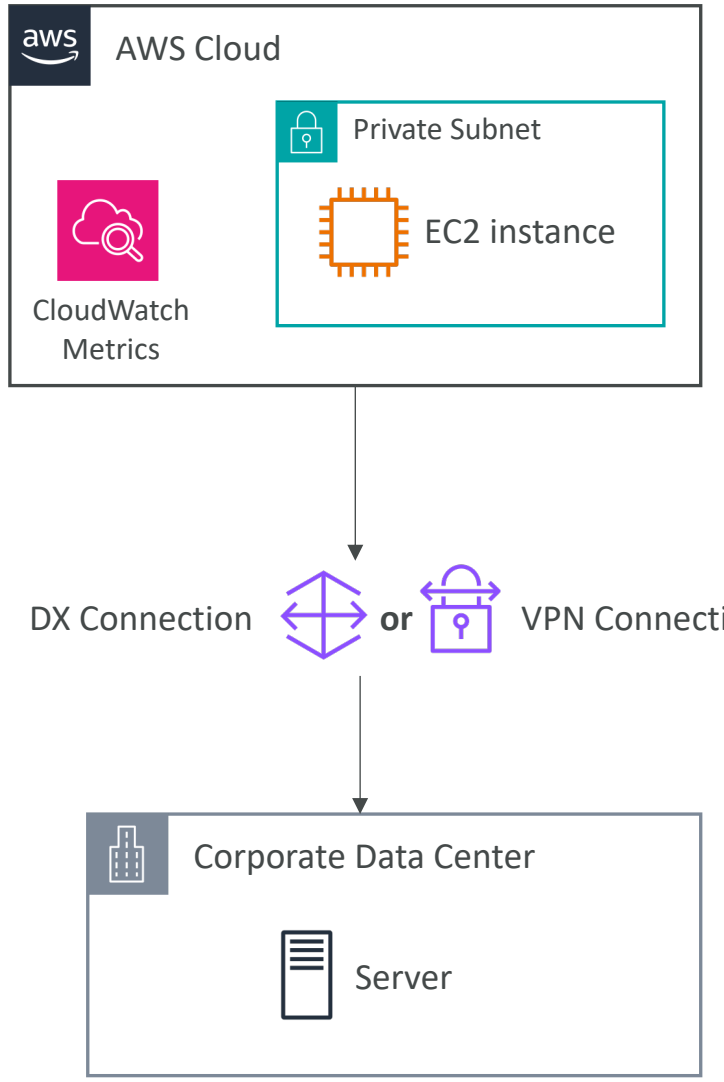
```
aws cloudwatch set-alarm-state --alarm-name "myalarm" --state-value
ALARM --state-reason "testing purposes"
```

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CloudWatch Network Synthetic Monitor

- Monitor and detects network issues between your apps hosted on AWS and your on-premises data center
- Identify any network performance degradation (e.g., packet loss, latency, jitter…)
- No agents required to be installed
- Tests ICMP or TCP traffic to IPv4/IPv4 on-premises destinations through **Direct Connect** or **S2S VPN** connections
- Publishes data to CloudWatch Metrics



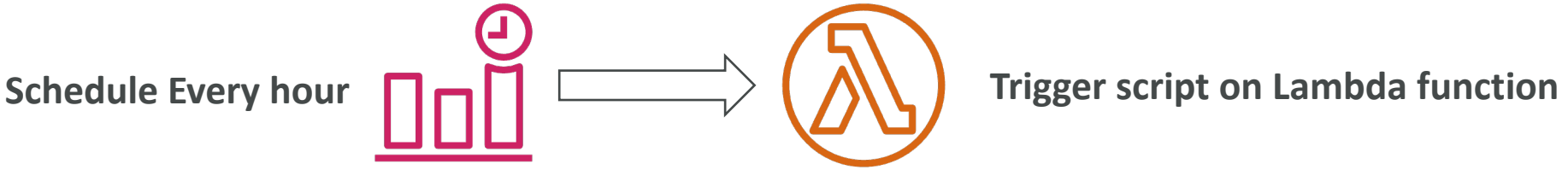
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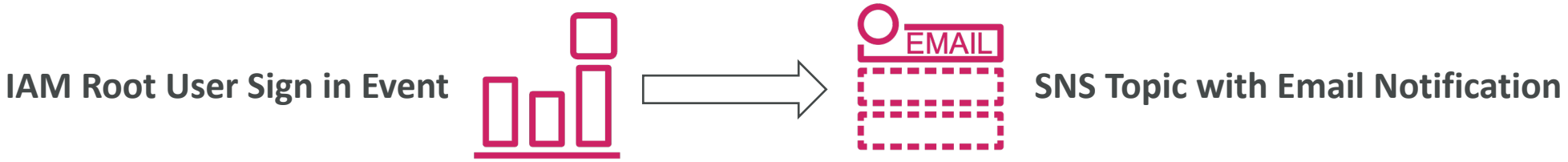
Amazon EventBridge (formerly CloudWatch Events)



- Schedule: Cron jobs (scheduled scripts)



- Event Pattern: Event rules to react to a service doing something

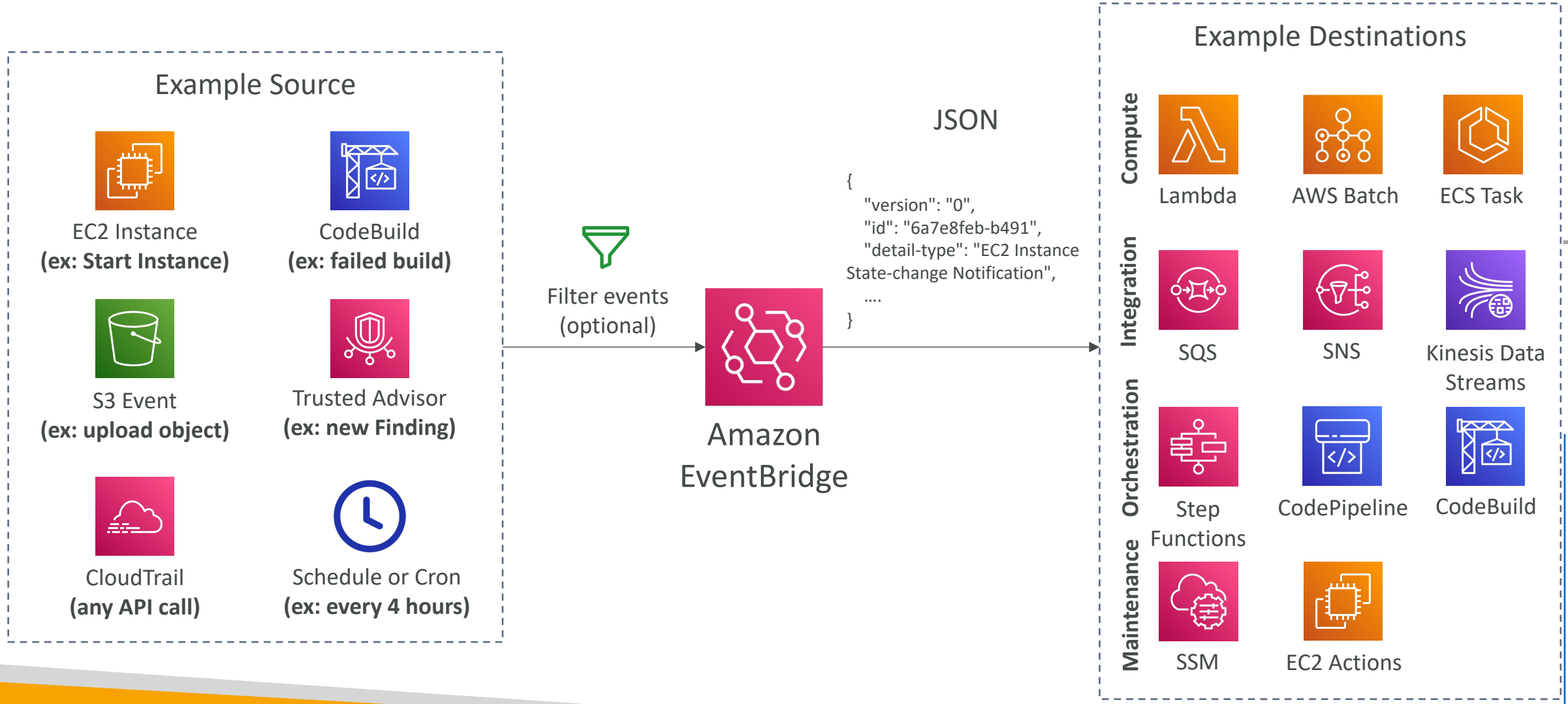


- Trigger Lambda functions, send SQS/SNS messages…

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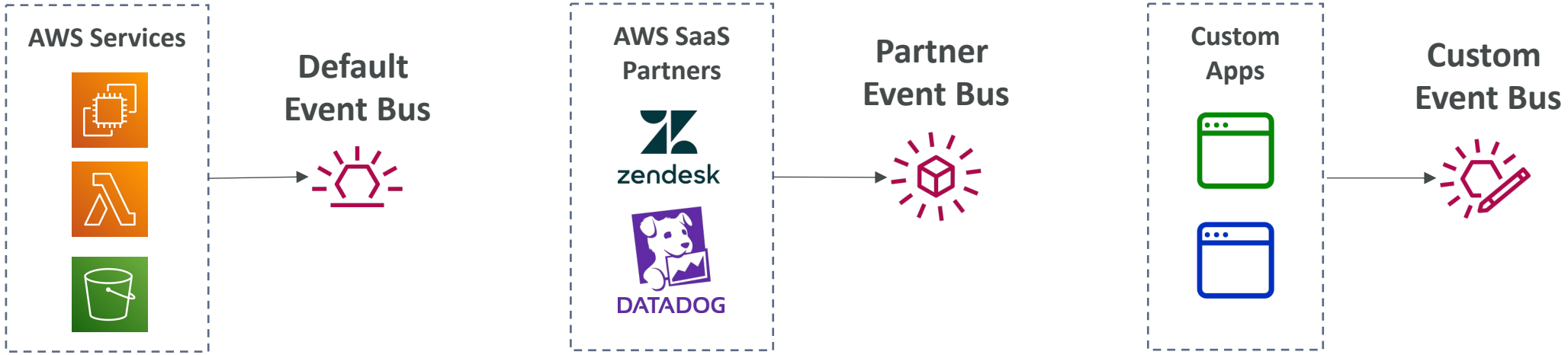
Amazon EventBridge Rules



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Amazon EventBridge



- Event buses can be accessed by other AWS accounts using Resource-based Policies

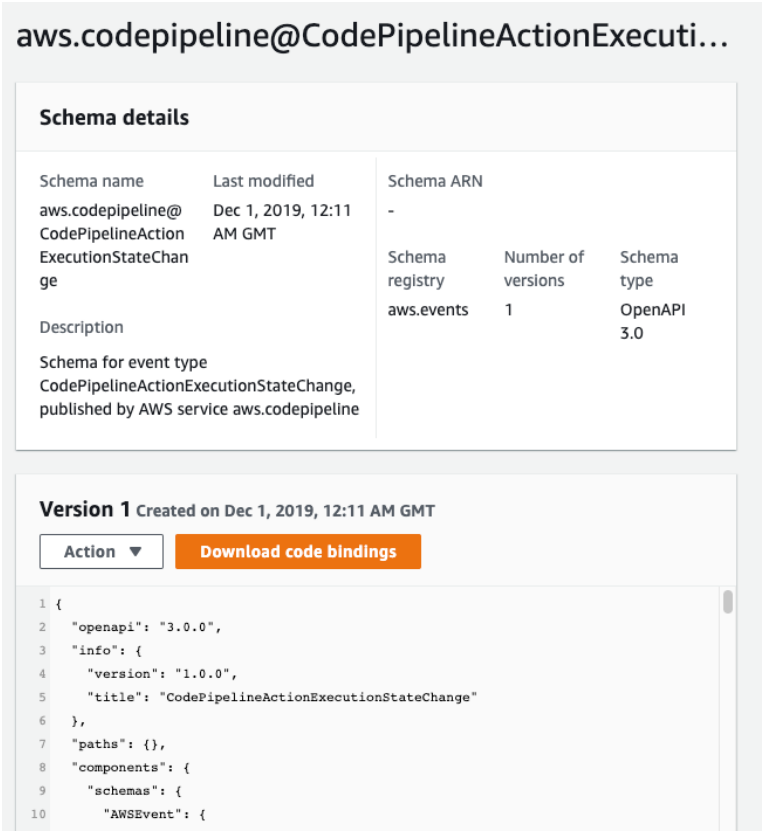
- You can **archive events** (all/filter) sent to an event bus (indefinitely or set period)
- Ability to **replay archived events**

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Amazon EventBridge – Schema Registry

- EventBridge can analyze the events in your bus and infer the **schema**
- The **Schema Registry** allows you to generate code for your application, that will know in advance how data is structured in the event bus
- Schema can be versioned



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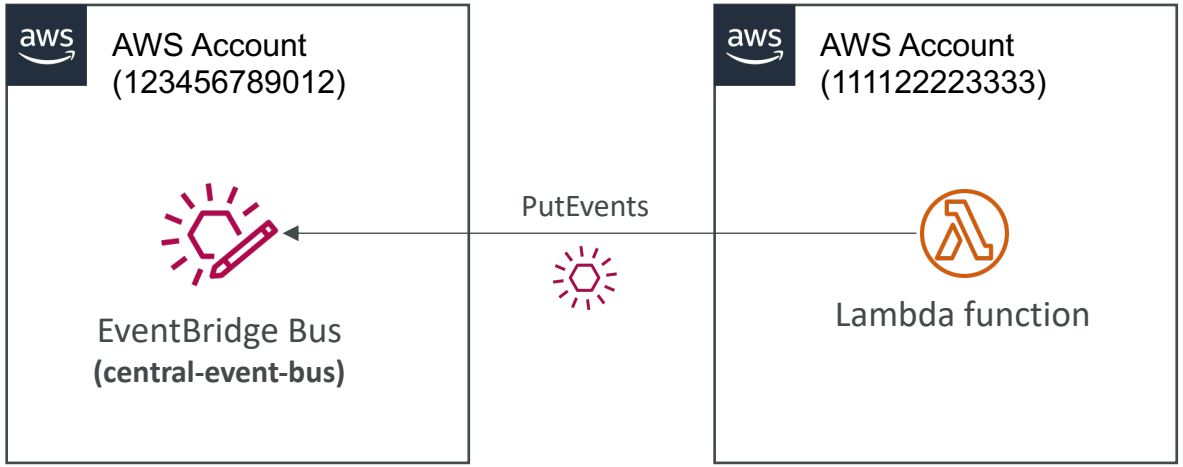
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Amazon EventBridge – Resource-based Policy

- Manage permissions for a specific Event Bus
- Example: allow/deny events from another AWS account or AWS region
- Use case: aggregate all events from your AWS Organization in a single AWS account or AWS region

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": "events:PutEvents",
            "Principal": { "AWS": "111122223333" },
            "Resource": "arn:aws:events:us-east-1:123456789012:
event-bus/central-event-bus"
        }
    ]
}
```

Allow **events** from another AWS account

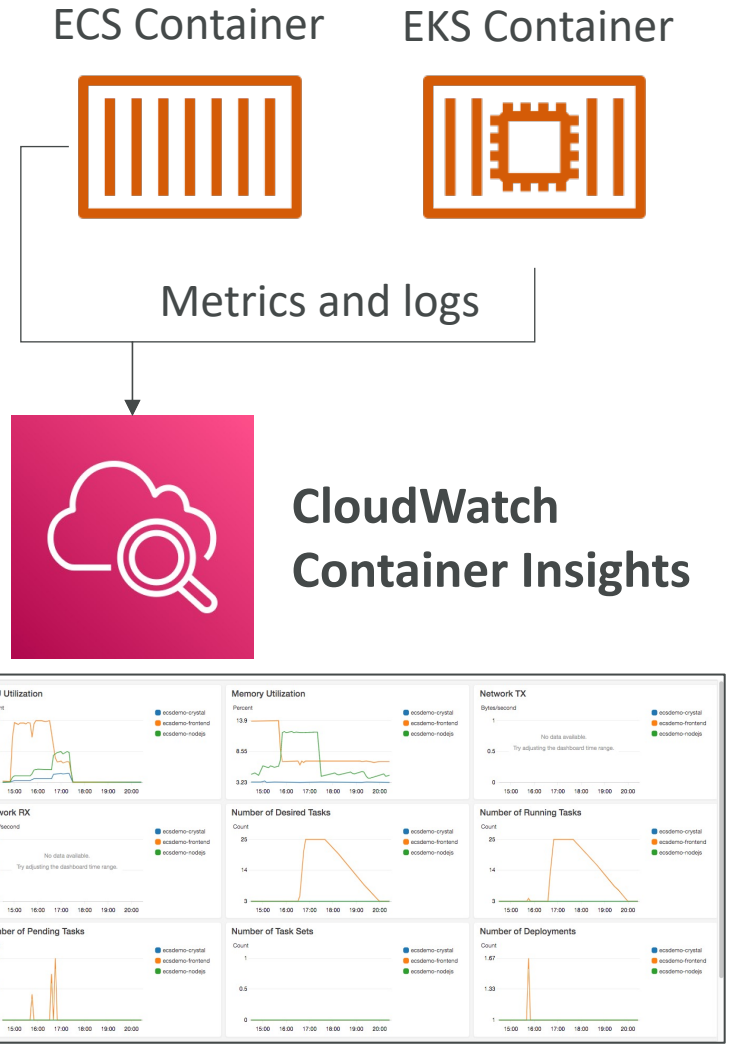


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CloudWatch Container Insights

- Collect, aggregate, summarize **metrics and logs** from containers
- Available for containers on…
  - Amazon Elastic Container Service (Amazon ECS)
  - Amazon Elastic Kubernetes Services (Amazon EKS)
  - Kubernetes platforms on EC2
  - Fargate (both for ECS and EKS)
- In Amazon EKS and Kubernetes, CloudWatch Insights is using a containerized version of the CloudWatch Agent to discover containers

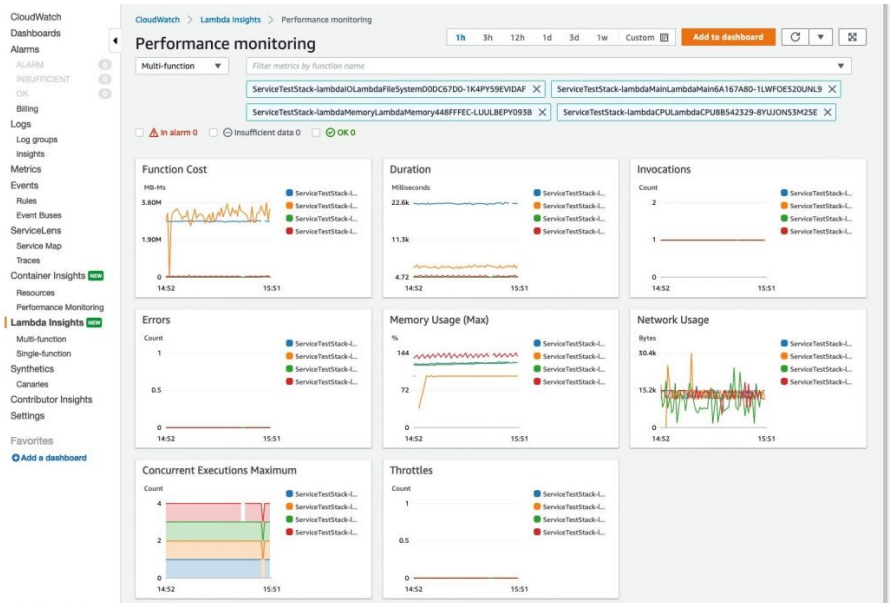


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CloudWatch Lambda Insights

- Monitoring and troubleshooting solution for serverless applications running on AWS Lambda
- Collects, aggregates, and summarizes system-level metrics including CPU time, memory, disk, and network
- Collects, aggregates, and summarizes diagnostic information such as cold starts and Lambda worker shutdowns
- Lambda Insights is provided as a Lambda Layer

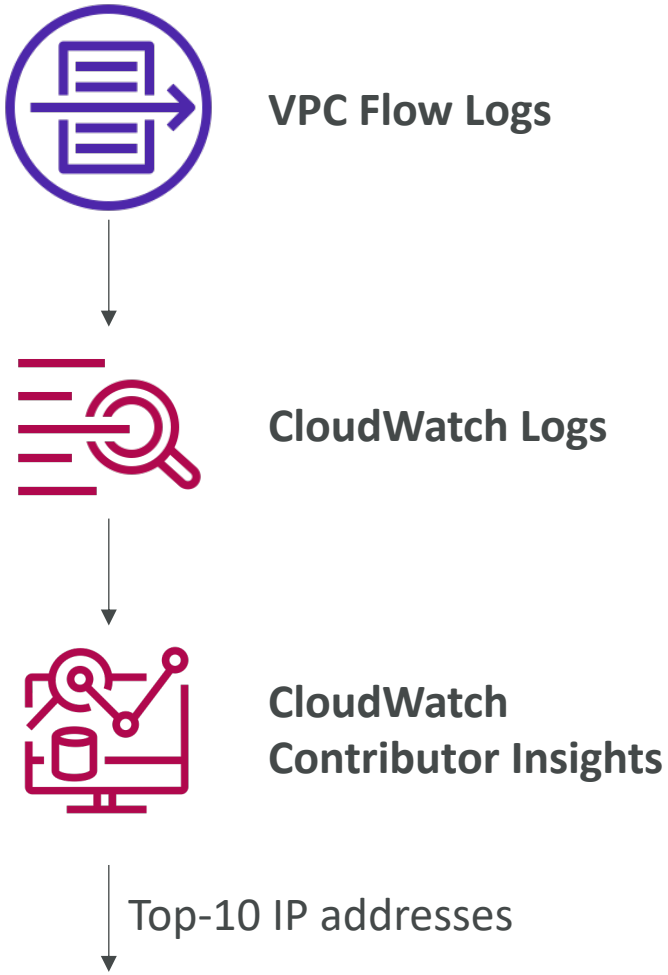


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CloudWatch Contributor Insights

- Analyze log data and create time series that display contributor data.
  - See metrics about the top-N contributors
  - The total number of unique contributors, and their usage.
- This helps you find top talkers and understand who or what is impacting system performance.
- Works for any AWS-generated logs (VPC, DNS, etc..)
- For example, you can find bad hosts, identify the heaviest network users, or find the URLs that generate the most errors.
- You can build your rules from scratch, or you can also use sample rules that AWS has created – leverages your CloudWatch Logs
- CloudWatch also provides built-in rules that you can use to analyze metrics from other AWS services.



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CloudWatch Application Insights

- Provides automated dashboards that show potential problems with monitored applications, to help isolate ongoing issues
- Your applications run on Amazon EC2 Instances with select technologies only (Java, .NET, Microsoft IIS Web Server, databases…)
- And you can use other AWS resources such as Amazon EBS, RDS, ELB, ASG, Lambda, SQS, DynamoDB, S3 bucket, ECS, EKS, SNS, API Gateway…

- Powered by SageMaker
- Enhanced visibility into your application health to reduce the time it will take you to troubleshoot and repair your applications
- Findings and alerts are sent to Amazon EventBridge and SSM OpsCenter

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CloudWatch Insights and Operational Visibility

- CloudWatch Container Insights
  - ECS, EKS, Kubernetes on EC2, Fargate, needs agent for Kubernetes
  - Metrics and logs
- CloudWatch Lambda Insights
  - Detailed metrics to troubleshoot serverless applications
- CloudWatch Contributors Insights
  - Find "Top-N" Contributors through CloudWatch Logs
- CloudWatch Application Insights
  - Automatic dashboard to troubleshoot your application and related AWS services

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AWS CloudTrail

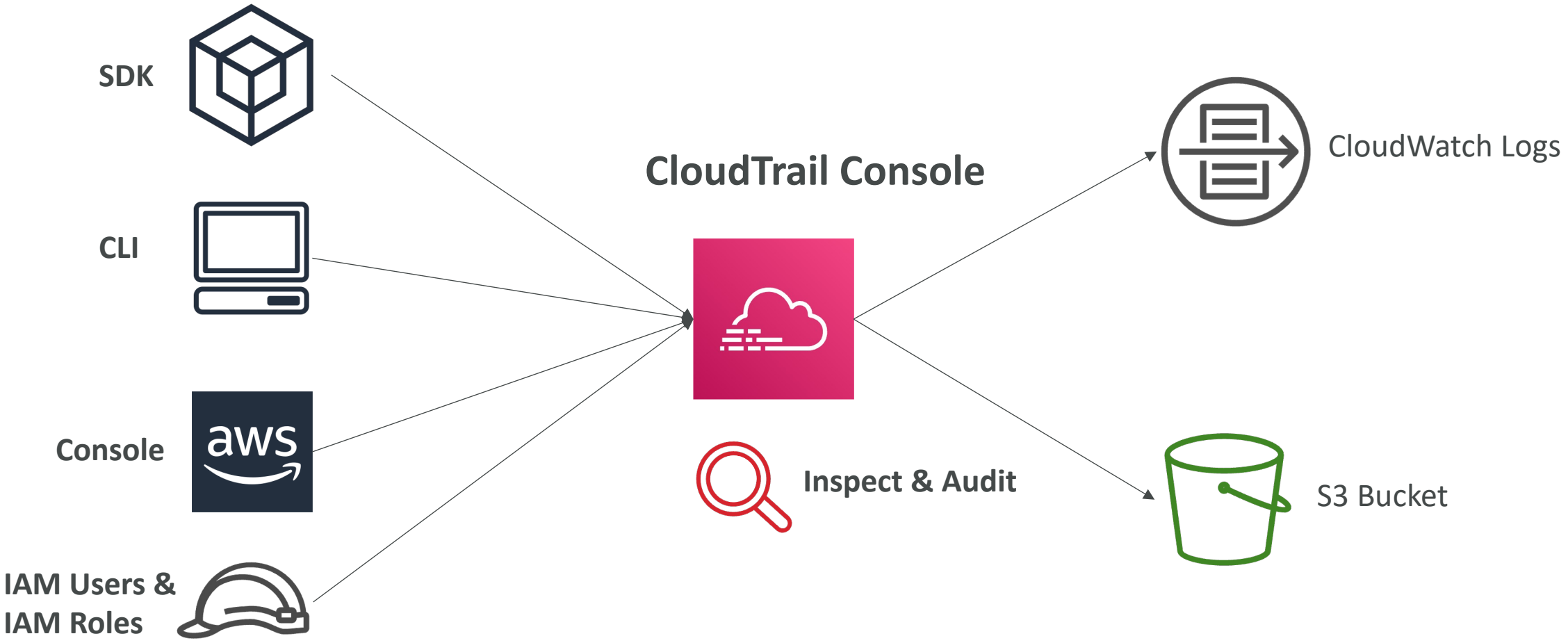


- Provides governance, compliance and audit for your AWS Account
- CloudTrail is enabled by default!
- Get an history of events / API calls made within your AWS Account by:
  - Console
  - SDK
  - CLI
  - AWS Services
- Can put logs from CloudTrail into CloudWatch Logs or S3
- A trail can be applied to All Regions (default) or a single Region.
- If a resource is deleted in AWS, investigate CloudTrail first!

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CloudTrail Diagram



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CloudTrail Events



- Management Events:
  - Operations that are performed on resources in your AWS account
  - Examples:
    - Configuring security (IAM **AttachRolePolicy**)
    - Configuring rules for routing data (Amazon EC2 **CreateSubnet**)
    - Setting up logging (AWS CloudTrail **CreateTrail**)
  - By default, trails are configured to log management events.
  - Can separate **Read Events** (that don't modify resources) from **Write Events** (that may modify resources)

- Data Events:
  - By default, data events are not logged (because high volume operations)
  - Amazon S3 object-level activity (ex: **GetObject**, **DeleteObject**, **PutObject**): can separate Read and Write Events
  - AWS Lambda function execution activity (the **Invoke** API)

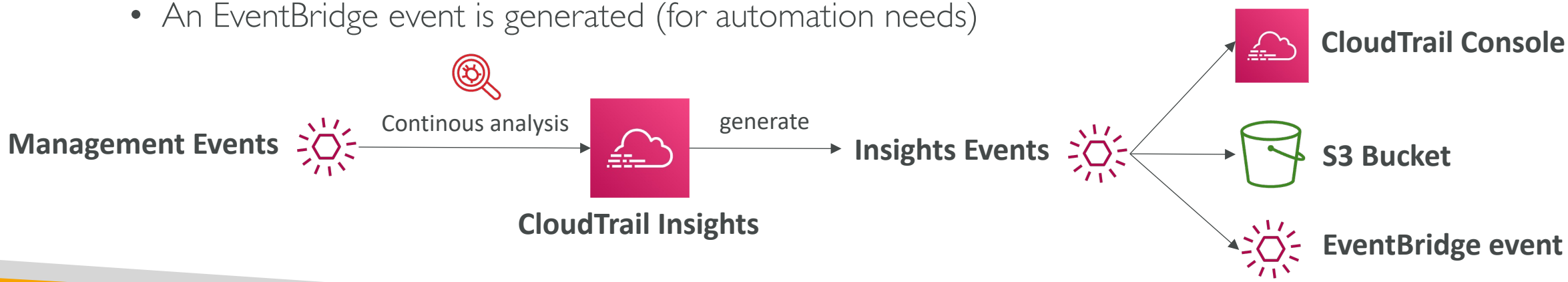
- CloudTrail Insights Events:
  - See next slide ☺

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CloudTrail Insights

- Enable **CloudTrail Insights to detect unusual activity** in your account:
  - inaccurate resource provisioning
  - hitting service limits
  - Bursts of AWS IAM actions
  - Gaps in periodic maintenance activity
- CloudTrail Insights analyzes normal management events to create a baseline
- And then **continuously analyzes write events to detect unusual patterns**
  - Anomalies appear in the CloudTrail console
  - Event is sent to Amazon S3
  - An EventBridge event is generated (for automation needs)

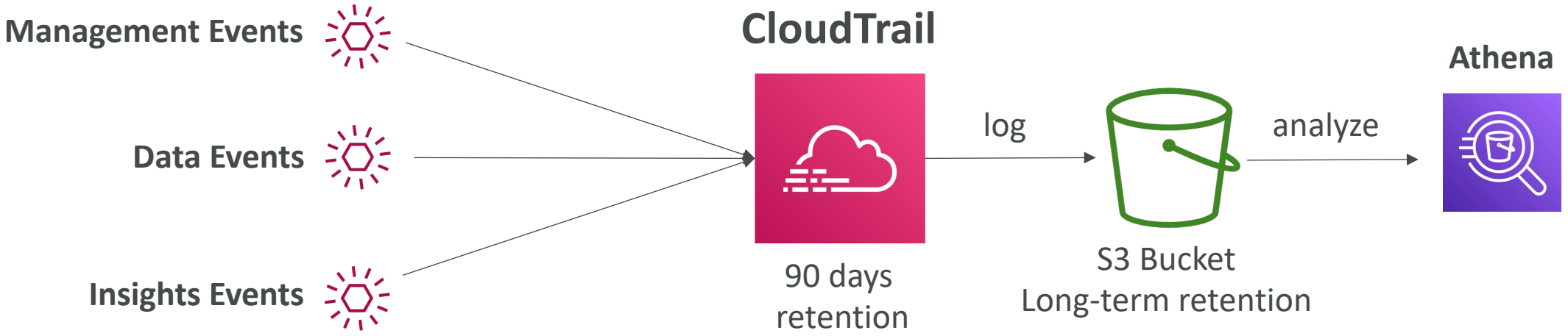


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CloudTrail Events Retention

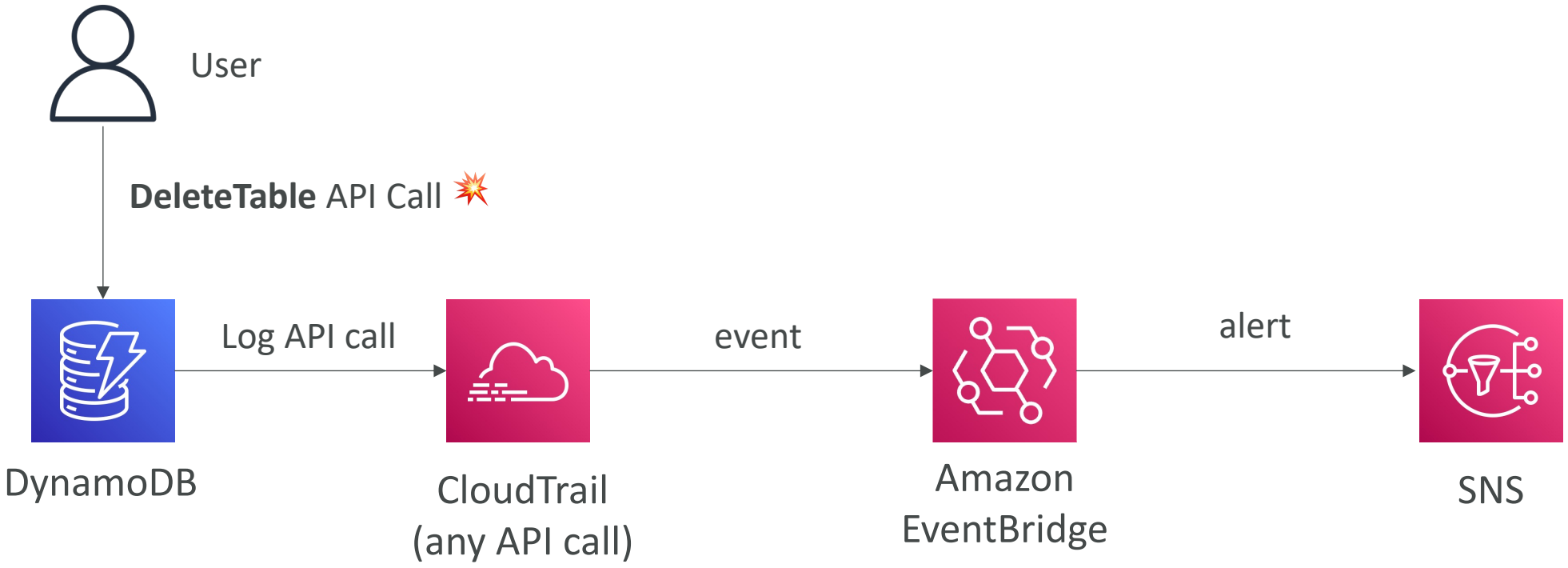
- Events are stored for 90 days in CloudTrail
- To keep events beyond this period, log them to S3 and use Athena



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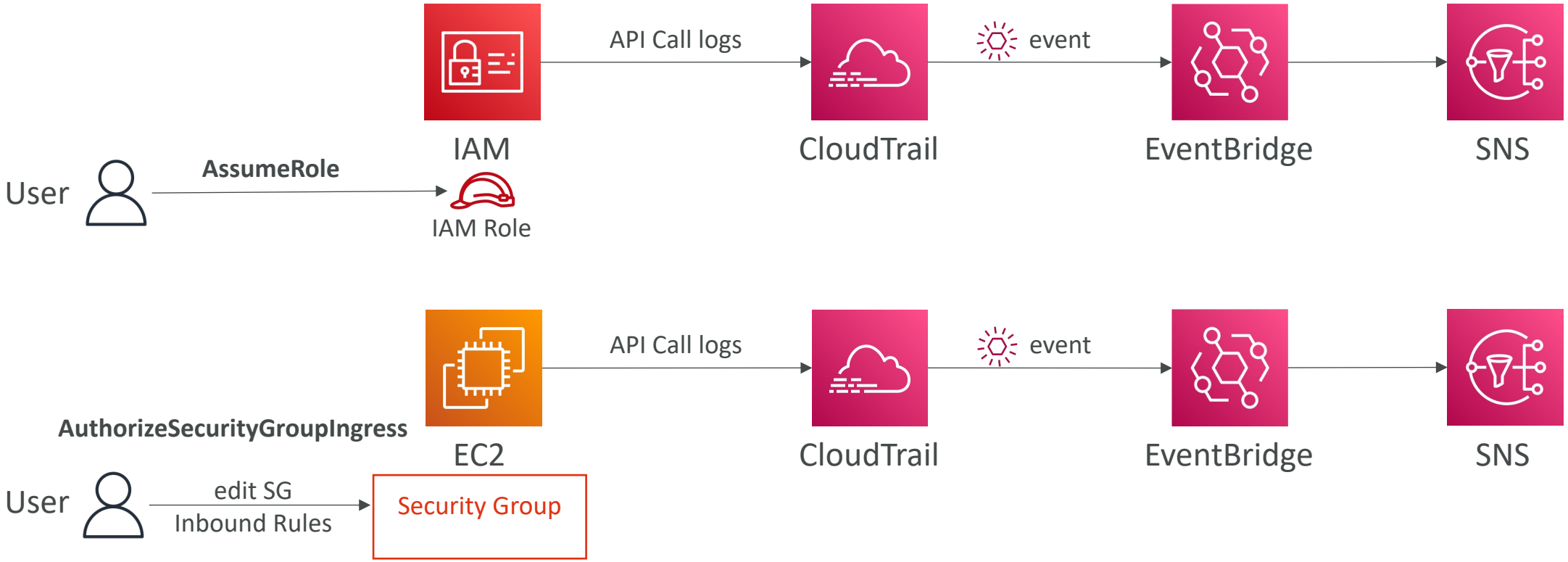
Amazon EventBridge – Intercept API Calls



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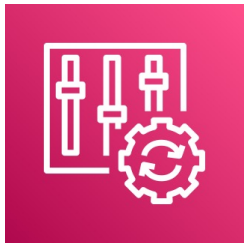
Amazon EventBridge + CloudTrail



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AWS Config



- Helps with auditing and recording **compliance** of your AWS resources
- Helps record configurations and changes over time
- Questions that can be solved by AWS Config:
  - Is there unrestricted SSH access to my security groups?
  - Do my buckets have any public access?
  - How has my ALB configuration changed over time?
- You can receive alerts (SNS notifications) for any changes
- AWS Config is a per-region service
- Can be aggregated across regions and accounts
- Possibility of storing the configuration data into S3 (analyzed by Athena)

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Config Rules

- Can use AWS managed config rules (over 75)
- Can make custom config rules (must be defined in AWS Lambda)
  - Ex: evaluate if each EBS disk is of type gp2
  - Ex: evaluate if each EC2 instance is t2.micro
- Rules can be evaluated / triggered:
  - For each config change
  - And / or: at regular time intervals
- **AWS Config Rules does not prevent actions from happening (no deny)**

- Pricing: no free tier, $0.003 per configuration item recorded per region, $0.001 per config rule evaluation per region

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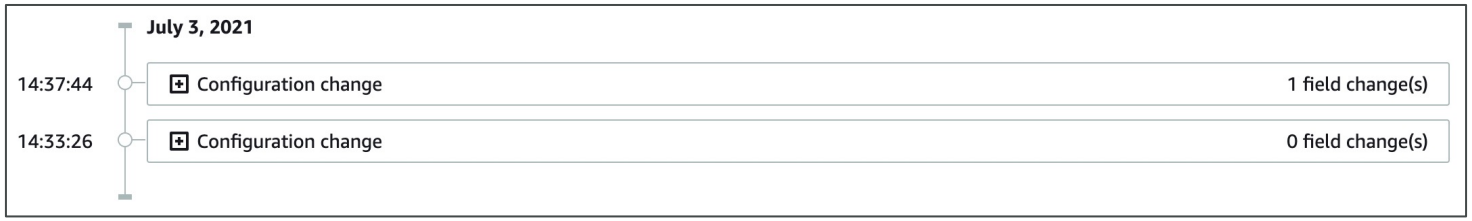
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AWS Config Resource

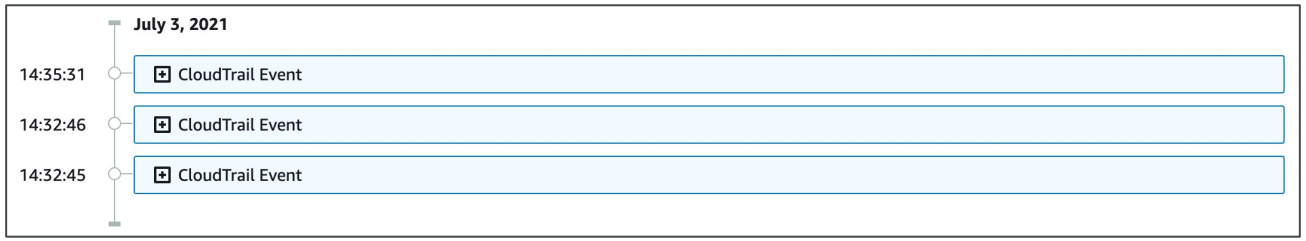
- View compliance of a resource over time



- View configuration of a resource over time



- View CloudTrail API calls of a resource over time

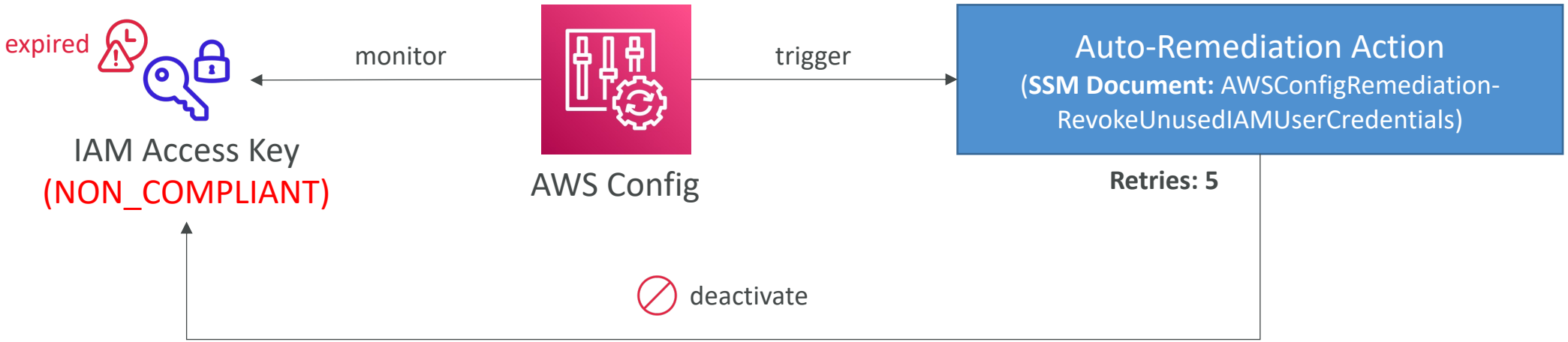


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Config Rules – Remediations

- Automate remediation of non-compliant resources using SSM Automation Documents
- Use AWS-Managed Automation Documents or create custom Automation Documents
  - Tip: you can create custom Automation Documents that invokes Lambda function
- You can set **Remediation Retries** if the resource is still non-compliant after auto-remediation

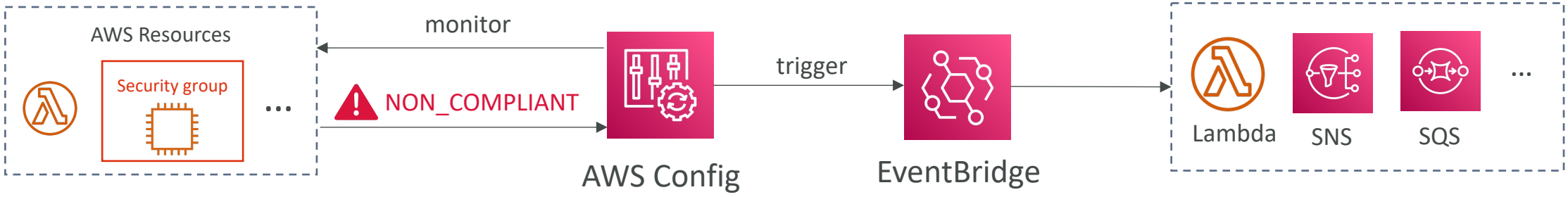


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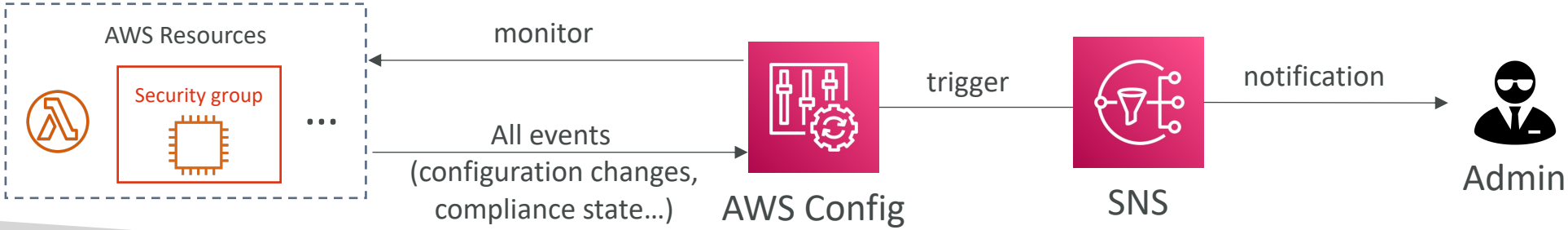
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Config Rules – Notifications

- Use EventBridge to trigger notifications when AWS resources are non-compliant



- Ability to send configuration changes and compliance state notifications to SNS (all events – use SNS Filtering or filter at client-side)



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CloudWatch vs CloudTrail vs Config

- CloudWatch
  - Performance monitoring (metrics, CPU, network, etc…) & dashboards
  - Events & Alerting
  - Log Aggregation & Analysis
- CloudTrail
  - Record API calls made within your Account by everyone
  - Can define trails for specific resources
  - Global Service
- Config
  - Record configuration changes
  - Evaluate resources against compliance rules
  - Get timeline of changes and compliance

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For an Elastic Load Balancer

- CloudWatch:
  - Monitoring Incoming connections metric
  - Visualize error codes as % over time
  - Make a dashboard to get an idea of your load balancer performance
- Config:
  - Track security group rules for the Load Balancer
  - Track configuration changes for the Load Balancer
  - Ensure an SSL certificate is always assigned to the Load Balancer (compliance)
- CloudTrail:
  - Track who made any changes to the Load Balancer with API calls

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